

Athulith Paraselli

GitHub: <https://github.com/aparaselli>
Email: pathulith@gmail.com
Portfolio: <https://aparaselli.github.io/>

RESEARCH INTERESTS

Trustworthy Machine Learning, Mechanistic Interpretability, Multimodal Artificial Intelligence

EDUCATION

Brown University, Providence, RI Sep 2025 - May 2027
Master of Science, Computer Science - Artificial Intelligence/Machine Learning GPA: 4.00 / 4.00

University of California, San Diego (UCSD), La Jolla, CA Sep 2021- June 2025
Bachelor of Science, Mathematics-Computer Science, *Data Science Minor* GPA: 3.82 / 4.00

RESEARCH EXPERIENCE

Graduate Research Assistant, Brown University Sep 2025 – Present
Advised by Dr. Ellie Pavlick

- Designed and conducted experiment to understand visual demographic biases within VLMs.
- Developed generalized pipeline for activation patching and logit lens within VLMs.
- Created steering vectors and applied them at inference to allow for training-free bias correction.

Undergraduate Research Assistant, UCSD Apr 2025 – Aug 2025
Advised by Dr. Xi Li and Dr. Zhuowen Tu

- Developed novel multimodal adversarial training pipeline to improve robustness of VLMs by 12%.
- Integrated LoRA and prompt tuning in a unified training loop, enabling flexible defense strategies.

Undergraduate Research Assistant, UCSD Nov 2024 – Nov 2025
Advised by Dr. George Sugihara

- Designed and implemented a hybrid-LSTM model in PyTorch, achieving 70% accuracy in forecasting algal blooms.
- Fine-tuned Chronos, a foundation model for time-series forecasting, on ecological data to reduce weighted quantile loss (wQL) of algal bloom predictions by 32%.

TECHNICAL EXPERIENCE

Software Engineer Intern, Qualcomm Jun 2024 – Sep 2024
Modem Integration Testing

- Created an efficient unit testing workflow through optimizing existing industry-standard methods.
- Implemented functionality to automatically generate unit tests based on customer specifications.

WORKING PAPERS

Predicting Bioluminescent Algal Blooms by Conditioning Neural Forecasts on Reconstructed Dynamics
Athulith Paraselli, Ciro Zhang, and George Sugihara

TEACHING EXPERIENCE

Undergraduate Instructional Assistant
Conduct exam review sessions, design homework/exams, hold weekly office hours, grade assignments

Principles of Data Science, UCSD Apr 2024 - Jun 2025
Advanced Machine Learning Methods, UCSD Apr 2025 - Jun 2025

TECHNICAL SKILLS

Programming Languages
Python, C/C++, Java, Perl, ARM Assembly, MATLAB, SQL, HTML, CSS, JavaScript, Svelte

Areas of Expertise
Version Control(Git/Github), Machine Learning(Scikit-learn, Pytorch, Tensorflow, Keras, Hugging Face, NNSight), AWS(Amplify, API Gateway, Lambda, DynamoDB)

SERVICES &
OUTREACH

Eta Kappa Nu (HKN) Projects Program

Nov 2024 – Jun 2025

Inaugural Project Lead

- Led a team of 4 students to develop a live data processing pipeline and website to display Bioluminescent algal bloom forecasts.
- Established connections between Labs at Scripps Institution of Oceanography and Eta Kappa Nu.
- Coordinated community outreach initiatives, presenting accessible explanations of Harmful Algal Blooms and sharing updates from ongoing research collaborations.
- Presented results at San Diego Undergrad Tech Conference.

Eta Kappa Nu (HKN) Events Team

Mar 2024 – Jun 2025

Math-CS Events Chair

- Organized and led technical workshops on fundamental ML concepts such as neural networks and gradient descent, enhancing accessibility for underclassmen.
- Collaborated with board members to coordinate events, recruit inductees, and promote technical engagement across the department.

AWARDS &
HONORS

- Eta Kappa Nu (IEEE HKN) Member, *UCSD*
- Provost Honors x11, *UCSD*

2023-2025

2021-2025